

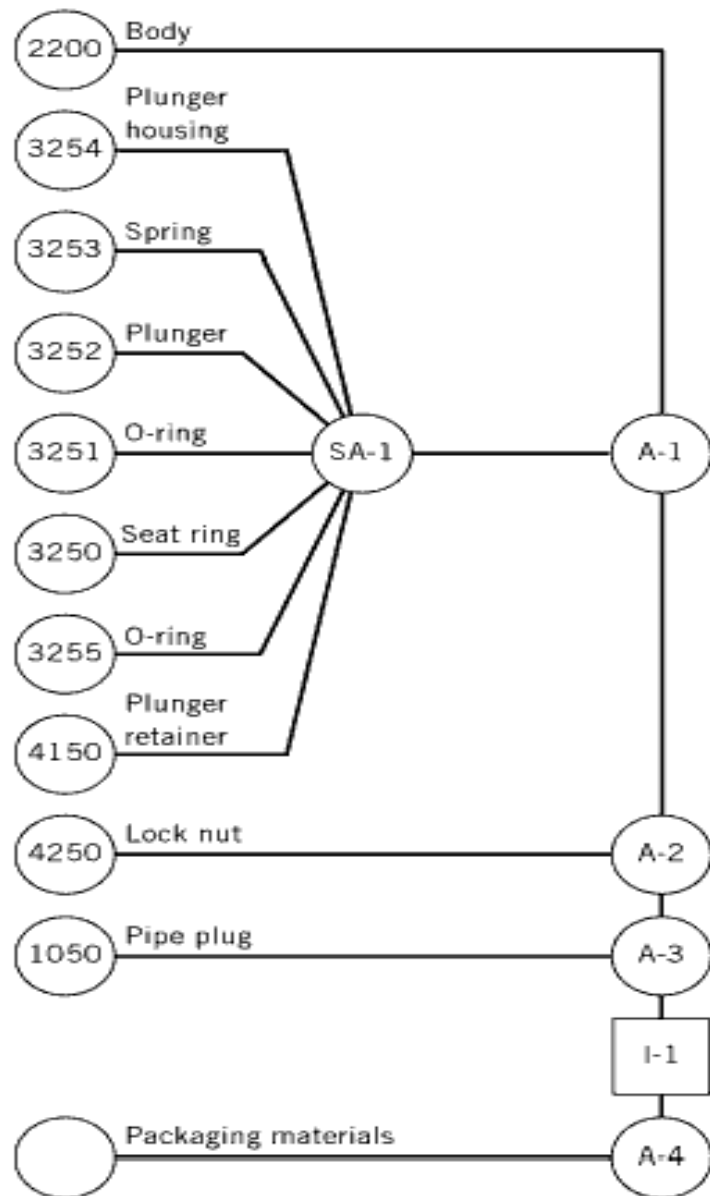
2.33. Sequencing Required Processes

- An assembly chart / drawing (It provides documentation for assembly). The easiest method of constructing an assembly chart is to begin with the completed product and to trace the product disassembly back to its basic components.

ROUTE SHEET

Company A.R.C., Inc. Part Name Plunger Housing Prepared by J.A.
 Produce Air Flow Regulator Part No. 3254 Date _____

Oper No.	Operation Description	Machine Type	Tooling	Dept.	Set-up Time (hr.)	Operation Time (hr.)	Materials or Parts Description
0104	Shape, drill, cut off	Automatic screw machine	.50 in. dia. collar, feed fingers, cir. form tool, .45 in. dia. center drill, .129 in. twist drill, finish spiral drill, cut off blade		5	.0057	Aluminum 1.0 in. dia. × 12 ft.
0204	Machine slot and thread	Chucker	.045 in. slot saw, turret slot attach. 3/8-32 thread chaser		2.25	.0067	
0304	Drill 8 holes	Auto. dr. unit (chucker)	.078 in. dia. twist drill		1.25	.0038	
0404	Deburr and blow out	Drill press	Deburring tool with pilot		.5	.0031	
SA1	Enclose subassembly	Dennison hyd. press	None		.25	.0100	



Assembly chart for an air flow regulator

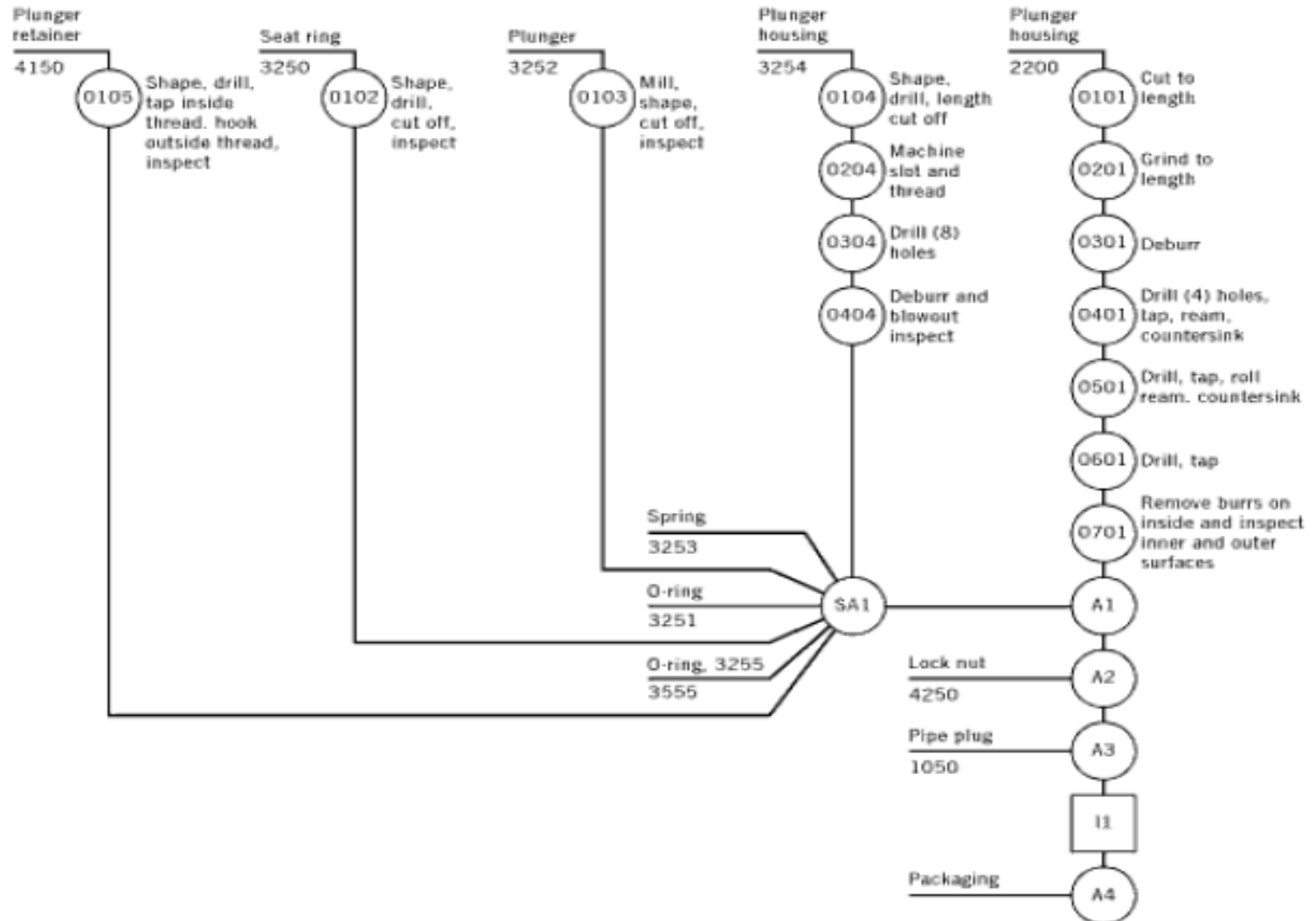
OPERATION PROCESS CHART

Company A.R.C., Inc.

Prepared by J. A.

Product Air Flow Regulator

Date _____



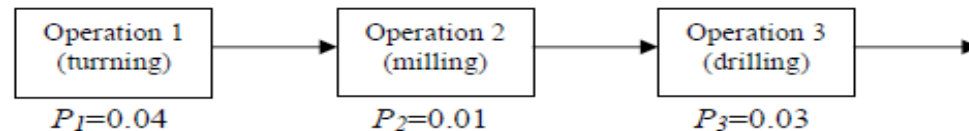
Operation process chart for the air flow regulator

2.4 Schedule Design

Schedule design provide answers to questions involving how much to produce and when to produce. Production quantity decisions are referred to as lot size decisions; determining when to produce is referred to as production scheduling. In addition to how much and when, it is important to know how long production will continue; such a determination is obtained from market forecasts.

- Market Information
- Process Requirements

A product has a market estimate of 97,000 components and requires three processing steps (turning, milling, and drilling) having scrap estimates of $P_1=0.04$, $P_2=0.01$, $P_3=0.03$. Calculate the production input to operation 1.



$$I_3 = \frac{97,000}{1 - 0.03} = 100,000$$

$$\left(I_k = \frac{O_k}{1 - P_k} \right)$$

$$I_2 = \frac{100,000}{1 - 0.01} = 101,010$$

$$I_1 = \frac{101,010}{1 - 0.04} = 105,219$$

$$\text{(or, } I_1 = \frac{97,000}{(1 - 0.04)(1 - 0.01)(1 - 0.03)} = 105,219 \text{)}$$

Summary of Production Requirements			
Operation	Production Quantity Scheduled (units)	Expected number of good units produced	
Turning	105,219	101,010	
Milling	101,010	100,000	
Drilling	100,000	97,000	

Table 4.5. Summary of production requirements

4.5. Facilities Design

The seven management and planning tools are the

1. **Affinity diagram,**
2. **The interrelationship digraph,**
3. **The tree diagram,**
4. **The matrix diagram,**
5. **The contingency diagram,**
6. **The activity network diagram, and**
7. **The prioritization matrix**